

## Response Letter with Annotations

Lu, X., Das, D., Huh, J., & Srivastava, J. (2022). *Influence of consumers' temporary affect on ad engagement: A computational research approach*. *Journal of Advertising*, 51(3), 352–368.  
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### Annotations

#### 1. Purposes

##### a. Motivation (one-sentence summary)

The motivation is to study the influence of consumers' temporary affective states on their selective attention to and engagement with different types of ads based on attention-grabbing characteristics.

##### b. Specific Rationale (several bullet points)

- Affect is a precursor of consumer responses to ad exposure.
- Mood management theory: Consumers select media content that helps to improve or maintain their current state.

##### c. Research Questions and/or Hypotheses

###### • Identify IV, DV, mediator, moderator

IV: Affective state (positive/negative/neutral)

DV: Four selective engagement with different types of ads (exciting/soothing, emotional/rational appeals, high/low semantic affinity, positive/negative)

Mediator: NA

Moderator: NA

###### • Identify specific relationships among variables

Summary: There is a matching relationship in all hypotheses that positive affect state is related to higher engagement when ad characteristics are exciting, emotional appeal, high semantic affinity, and positive than negative affect state.

H1: Consumers in the positive affective state will have greater tendency to be drawn to engage with more exciting ads than those in the negative affective state.

H2: Consumers in the negative affective state will have greater tendency to be drawn to engage with ads using emotional appeals than those in the positive affective state.

H3: Consumers in the positive affective state will likely be drawn to engage with ads with high semantic affinity with the prior event generating the affective state, whereas those in the negative affective state will likely be drawn to engage with ads with lower semantic affinity.

H4: Consumers in the negative affective state will have greater tendency to be drawn to engage with positively valenced ads than those in the positive affective state.

#### 2. Method

**a. Research Design**

Quasi experiment using TV content data (game and ads) and Twitter's tweet data (consumers)

Experimental condition for affective state: (1) positive, (2) neutral, and (3) negative affective state

**b. Sample (representative sample?)**

Sample is users affiliated with New England Patriots or the Los Angeles Rams  
Not representative for general population

**c. Measures**

Affective state: Whether users' affiliated team scored in the game was coded

Types of ads: Excitatory potential, emotional appeal, semantic affinity, and valence was coded by two coders.

Selective engagement: They developed a proxy measure: they first identified tweets about ads and make a match of each type of ads; they compared relative numbers of tweets to compare engagement for different types of ads.

**d. Analysis**

They used a hierarchical logistic regression analysis.

**3. Results and Conclusions**

**a. Answers to RQs and Hypotheses**

H1 was not supported: affective state did not significantly influence engagement with exciting or soothing ads.

H2 was not supported: affective state did not significantly influence engagement with emotional or rational ads.

H3 was supported: consumers in a positive affective state would more likely be drawn to engage with high semantic-affinity ads compared to those in a negative affective state.

H4 was supported: consumers in a negative affective state would more likely be drawn to engage with positively valenced ads compared to those in a positive affective state.

**b. Include Key Figures/Tables**

**Table 2.** Hierarchical logistic regression for affective state and engagement with exciting versus soothing ads.

| Predictors                  | Model 1                                             |         |           | Model 2                                             |         |        |
|-----------------------------|-----------------------------------------------------|---------|-----------|-----------------------------------------------------|---------|--------|
|                             | B                                                   | SE      | Wald      | B                                                   | SE      | Wald   |
| Use of celebrity            | 2.28***                                             | .19     | 138.02    | 2.28***                                             | .19     | 137.86 |
| Inclusion of novel elements | .82***                                              | .22     | 13.46     | .84***                                              | .22     | 13.95  |
| Use of music                |                                                     |         | 174.74*** |                                                     |         | 175.07 |
| Use of music (1)            | -20.22                                              | 2618.41 | .00       | -20.24***                                           | 2613.41 | .00    |
| Use of music (2)            | -17.77                                              | 2618.41 | .00       | -17.76                                              | 2613.41 | .00    |
| Loud sound                  | -19.35                                              | 2618.41 | .00       | -19.35                                              | 2613.41 | .00    |
| Production pacing           | .35                                                 | .25     | 1.84      | .35                                                 | .26     | 1.89   |
| Affective state             |                                                     |         |           | -.22                                                | .19     | 1.41   |
| Cox and Snell $R^2$         |                                                     | .36     |           |                                                     | .36     |        |
| Model statistics            | $-2LL = 1045.72; \chi^2 = 783.90; df = 6; p < .001$ |         |           | $-2LL = 1044.32; \chi^2 = 785.30; df = 7; p < .001$ |         |        |

Note: Use of music consists of three values and was dummy-coded; \*\*\* $p < .001$ .

**Table 3.** Hierarchical logistic regression for affective state and engagement with emotional versus rational ads.

| Predictors                  | Model 1                                                   |         |           | Model 2                                                   |         |          |
|-----------------------------|-----------------------------------------------------------|---------|-----------|-----------------------------------------------------------|---------|----------|
|                             | B                                                         | SE      | Wald      | B                                                         | SE      | Wald     |
| Use of celebrity            | .58***                                                    | .15     | 14.15     | .59***                                                    | .15     | 14.73    |
| Inclusion of novel elements | −1.60***                                                  | .17     | 84.27     | −1.61***                                                  | .17     | 85.50    |
| Use of music                |                                                           |         | 20.184*** |                                                           |         | 21.22*** |
| Use of music (1)            | −19.32                                                    | 2367.77 | .00       | −19.32                                                    | 2366.25 | .00      |
| Use of music (2)            | −20.02                                                    | 2367.77 | .00       | −20.04                                                    | 2366.25 | .00      |
| Loud sound                  | −23.68                                                    | 2367.77 | .00       | −23.72                                                    | 2366.25 | .00      |
| Production pacing           | 4.16***                                                   | .32     | 166.43    | 4.16***                                                   | .32     | 166.56   |
| Affective state             |                                                           |         |           | .239                                                      | .155    | 2.38     |
| Cox and Snell $R^2$         |                                                           | .33     |           |                                                           | .33     |          |
| Model statistics            | −2LL = 1432.60; $\chi^2 = 681.84$ ; $df = 6$ ; $p < .001$ |         |           | −2LL = 1430.19; $\chi^2 = 684.25$ ; $df = 7$ ; $p < .001$ |         |          |

Note: Use of music consists of three values and was dummy-coded; \*\*\* $p < .001$ .

**Table 4.** Hierarchical logistic regression for affective state and engagement with high versus low semantic-affinity ads.

| Predictors                  | Model 1                                                   |         |          | Model 2                                                    |         |          |
|-----------------------------|-----------------------------------------------------------|---------|----------|------------------------------------------------------------|---------|----------|
|                             | B                                                         | SE      | Wald     | B                                                          | SE      | Wald     |
| Use of celebrity            | 1.49***                                                   | .23     | 42.56    | 1.50***                                                    | .23     | 43.12    |
| Inclusion of novel elements | −2.60***                                                  | .20     | 177.99   | −2.63***                                                   | .20     | 178.61   |
| Use of music                |                                                           |         | 17.40*** |                                                            |         | 17.20*** |
| Use of music (1)            | −19.15                                                    | 5151.61 | .00      | −19.18                                                     | 5139.79 | .00      |
| Use of music (2)            | −14.92                                                    | 5151.61 | .00      | −14.98                                                     | 5139.79 | .00      |
| Loud sound                  | .76**                                                     | .28     | 7.34     | .73**                                                      | .28     | 6.61     |
| Production pacing           | 4.00***                                                   | 1.03    | 15.01    | 3.99***                                                    | 1.03    | 14.98    |
| Affective state             |                                                           |         |          | .40**                                                      | .19     | 4.27     |
| Cox and Snell $R^2$         |                                                           | .49     |          |                                                            | .49     |          |
| Model statistics            | −2LL = 993.65; $\chi^2 = 1171.36$ ; $df = 6$ ; $p < .001$ |         |          | −2LL = 989.29; $\chi^2 = 1175.72$ ; $df = 7$ ; $p < .0011$ |         |          |

Note: Use of music consists of three values and was dummy-coded; \*\*\* $p < .001$ ; \*\* $p < .05$ .

### c. Authors' Conclusions

Researchers conclude that consumers' temporary affective states can influence selective engagement. The significant findings related to semantic affinity and valence are consistent with mood management theory. The insignificant results related to excitatory potential and emotional appeal may be due to their complex and subtle features, which are difficult to isolate in a naturalistic setting.

## 4. Discussion

### a. Implications

Researchers highlight three theoretical implications. This research contributes to the ad literature that consumers' affective states. It also extends the application of mood management theories in ad research. The study offers a novel perspective to study this topic by applying computation method. Researchers also include practical implications for recommender systems, suggesting that consumers' affective states can be used to offer personalized ads for more effective advertising strategies.

### b. Limitations

The findings may not be generalized for other sporting events. Sample also has limitations because Twitter data do not fully represent the broader Super Bowl viewer population. There is a limitation in developing proxy measures.

Computational methods also involve some confounding factors that could affect results.

**c. Future Directions**

Future directions follow from the limitations above. Proxy measures need to be further refined. Future studies could employ triangulation to mitigate the limitations of relying on a single method.

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**Response Letter (~600 words)**

- **Your assessment of the study**

The study focuses on influence of consumers' temporary affective states on their selective attention to and engagement with different types of ads based on attention-grabbing characteristics. Researchers categorize ads along four mood-relevant dimensions derived from mood management theory. They apply computational methods to study this research question based on Super Bowl data. It's highly relevant to computational method in advertising area and focuses on consumer-affect research.

Regarding strengths, its core idea is relatively innovative: a matching perspective between consumers' feature and engagement with different types of ad features is highly aligned with real-world media settings. It also fills an important gap in the literature, likely because such a design is difficult to execute using traditional experimental methods (e.g., too many experimental conditions). Theoretically, this research makes a compelling case for applying mood management theory that how media content is selected by people could be applied in a digital advertising context. This application is well-justified and can explain their hypotheses. In terms of method, I especially like their ideas of applying a computational method in studying consumers' affective state instead of a common self-reported measure. This approach really increases ecological validity and is very inspiring for future ad research. Their proxy measures for selective engagement and affective state are thoughtfully constructed and transparently discussed, and although the authors acknowledge their limitations, the overall approach is both creative and methodologically sound. Also, authors did an excellent job translating complex research into a clear and engaging journal article. Despite the technical nature of the computational approach, the authors maintain clarity by explaining key concepts without overwhelming the reader.

Regarding weakness, research with a single study applying computational methods may introduce confounding factors, which could be resolved by adding one additional study such as an experiment in a more controlled environment. Similar to authors' opinions, triangulation could help address this limitation. Another point that could be further developed is the interpretation of the non-significant findings for excitatory potential and emotional appeal. I think these results may indicate boundary conditions for mood management theory in advertising context, and additional discussion on this possibility would strengthen the theoretical contribution. Moreover, as authors discuss in their limitations, the Super Bowl might be a unique event that may not reflect other media environments. It seems that Super Bowl is a very common choice in marketing research (based on my readings from different classes). However, the Super Bowl's unusually high emotional intensity and cultural significance may limit the extent to which these findings can be generalized to more typical viewing situations. This might also provide another perspective on why results related to excitatory potential and emotional appeal are not significant.

In terms of computational methods, although the authors frame their approach as computational, the study still relies on manual coding, especially for the ad-level variables such as semantic affinity, emotional appeal, excitatory potential, and valence. While the data collection and matching rely on computational methods, the interpretation of ad characteristics remains human-driven. I think there may be more efficient ways to handle this to reduce the time required (e.g., LIWC or other ways).

Authors make contributions to connect new with old. This article addresses multiple theoretical contributions to both relevant literature and theoretical foundations. The study extends mood management theory into the advertising context and identifies which mood-relevant ad features matter most in naturalistic settings. Moreover, the research also offers insights that could be highly valuable for ad practitioners and recommender systems to develop personalized strategies. They could consider incorporating mood-relevant features from customers to offer personalization.